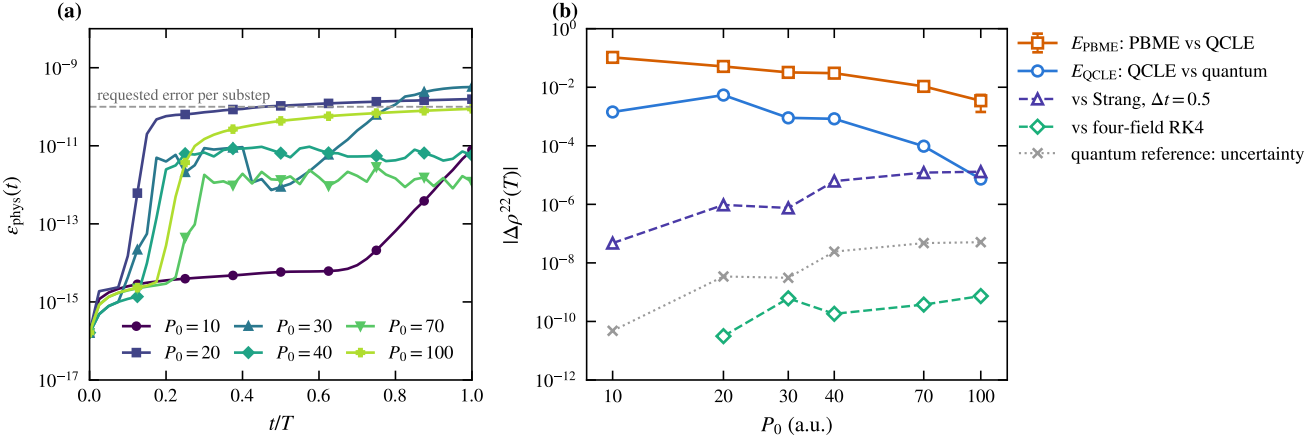
(a)
requested error per substep
10^{-9}
10^{-11}
10^{-13}
10^{-15}
10^{-17}
$\epsilon_{\mathrm{phys}}(t)$
$P_0 = 10$
$P_0 = 20$
$P_0 = 30$
$P_0 = 40$
$P_0 = 70$
$P_0 = 100$
0.0
0.2
0.4
0.6
0.8
1.0
t/T
(b)
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
$|\Delta\rho^{22}(T)|$
10
20
30
40
70
100
P_0 (a.u.)
E_{PBME}: PBME vs QCLE
E_{QCLE}: QCLE vs quantum
vs Strang, $\Delta t = 0.5$
vs four-field RK4
quantum reference: uncertainty